

PAPER

Autoregressive ConvNeXt-transformer fusion framework for polyphonic optical music recognition with focal loss optimization

Jun Wu^{1,*} and Wanshan Guo²

¹*School of Music and Dance, Chongqing Preschool Education College, Chongqing, China*

²*School of Big Data and Software Engineering, Chongqing University, Chongqing, China*

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Abstract: Optical music recognition technology has significantly enhanced the efficiency and accuracy of computational score transcription through deep learning methodologies. Although current techniques demonstrate strong performance in processing monophonic and single-voice scores, they struggle to achieve comparable accuracy when handling complex scores containing harmonic intervals, chords, polyphony, or multivoice compositions. In this paper, we propose ConvNeXt-Transformer Fusion (CNTF), an autoregressive end-to-end neural network framework employing an image-to-sequence architecture specifically optimized for automated transcription of intricate musical scores. The model integrates a ConvNeXt-based encoder for sheet music feature extraction and a Transformer-based decoder that generates transcription sequences through autoregressive prediction. To address class imbalance during training, we implement Focal Loss optimization. Experimental results demonstrate that the CNTF model achieves state-of-the-art performance in polyphony-rich score recognition, exhibiting superior character, symbol, and line error rates to existing baseline systems.

Keywords: Transformer networks, ConvNeXt backbone, CNTF framework, Focal loss optimization, Optical sheet music recognition, End-to-end (E2E) architecture

1. INTRODUCTION

Music serves as both a vital medium for emotional expression and a crucial conduit for cultural exchange [1–3]. Its influence extends far beyond artistic domains (e.g., cinema, choreography, television, and theatrical productions) into healthcare, sports, philosophy, psychology, education, and computer science, profoundly shaping human existence and the perpetuation of civilizational heritage. As a fundamental medium for musical documentation and transmission, sheet music systematically encodes compositional elements—including genre, thematic material, melodic contours, and rhythmic patterns—through standardized notational conventions. However, conventional music notation practices rely on manual transcription, with dissemination and preservation constrained by physical sheet music circulation and storage. Although culturally significant, this analog methodology is labor-intensive, resource-consuming, and vulnerable to variability introduced by human expertise and environ-

mental contingencies [4–7]. Consequently, advancing intelligent optical music recognition (OMR) systems—prioritizing efficiency, precision, and accessibility—holds great potential for facilitating musical practices and providing a broader interdisciplinary impact.

With the advent of the digital age, OMR has garnered growing attention across musicology, digital humanities, and computational arts [8–11]. OMR refers to computational methods for converting printed sheet music into machine-readable formats, thereby enabling applications in music pedagogy, score digitization, musicological analysis, copyright verification, and cultural heritage preservation. As a foundation for advanced music intelligence systems, OMR also supports automated composition, adaptive recommendation, and generative music technologies. OMR refers to a computational method for digitizing printed sheet music into machine-readable formats. This technology enables multidisciplinary research in musicology, digital humanities, and computational arts, with transformative applications in music-assisted pedagogy, notation tracking, automated score digitization, musicological analysis, software development, the digital preservation of cultural heritage, and copyright authentication.

*e-mail: 18223799668@163.com
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As a foundational component of advanced music intelligence systems, OMR provides critical infrastructure for automated composition, adaptive recommendation systems, and generative music technologies. Techniques integral to OMR—such as feature matching, image enhancement, and symbolic pattern recognition—are equally vital in industrial automation, manufacturing instrumentation, and multimedia informatics [12–15].

Conventional OMR systems generally rely on multi-stage pipelines that include image preprocessing [16,17], symbol recognition [18–20], musical information reconstruction [21,22], and symbolic encoding. These modular approaches require multiple isolated steps, making them complex and time-consuming. Recent advances in artificial intelligence have improved the efficiency of such pipelines and enabled the accurate recognition of simple monophonic scores, substantially reducing workflow complexity compared with earlier methods [23–26]. Nevertheless, these improvements still operate within multistage frameworks, which has motivated the development of E2E architectures that directly map score images to symbolic sequences. A representative example is the convolutional recurrent neural network (CRNN), in which CNNs extract visual features and RNNs decode sequential semantics [27]. Whereas E2E models simplify the pipeline and achieve strong performance, CRNNs remain constrained in capturing complex musical structures such as harmonic intervals, block chords, and polyphony, despite their high accuracy on monophonic tasks.

To overcome these challenges, Transformer-based architectures have recently been introduced: TrOMR [28] enhances global perceptual capability through a consistency loss and tailored data annotation strategy, improving recognition accuracy in real-world scores, whereas the sheet music transformer (SMT) [29] is the first E2E OMR model explicitly designed for polyphonic transcription, directly mapping score images into standard symbolic encodings without relying on monophonic simplifications.

Building on this context, we propose CNTF, an autoregressive image-to-sequence architecture that combines ConvNeXt-based hierarchical feature extraction with a Transformer-based decoder for symbolic sequence generation. Unlike previous E2E OMR models such as TrOMR [28] and SMT [29], CNTF is specifically designed to enhance polyphonic transcription performance while addressing symbol imbalance through Focal Loss optimization.

The key contributions of this paper are as follows:

We propose CNTF, a novel E2E OMR framework that bridges visual and symbolic modalities by integrating ConvNeXt and Transformer components.

We address severe symbol imbalance with Focal Loss, improving robustness in recognizing rare musical elements

(e.g., grace notes and fermata).

We compare our framework by experiments with existing frameworks and our framework demonstrates consistent improvements across multiple evaluation metrics (e.g., character and symbol error rates).

We analyze the limitations of CNTF (e.g., sensitivity to input quality and computational complexity) and propose strategies for further optimization.

2. RELATED WORK

In addition to mainstream OMR approaches, several complementary research directions have been explored. Human-in-the-loop and adaptive systems, for instance, have demonstrated the high-precision transcription of polyphonic Western scores by incorporating symbolic music data generation and iterative human corrections [30,31]. Multimodal and hybrid frameworks combine image enhancement, note recognition, print layout identification, and interactive user interfaces, enabling robust automated score digitization in networked environments [32,33]. Novel neural architectures, such as spiking neural networks (SNNs) and CNN-based models for position recognition (PR) and duration recognition (DR), have achieved accurate feature detection under both noisy and controlled conditions, even outside E2E OMR pipelines [34,35]. Transformer-based extensions, including sequence-to-sequence frameworks with mask language modeling, have addressed limitations in training efficiency and sequence accuracy while optimizing evaluation metrics for digital and nonstandard notation [36,37]. Finally, cross-disciplinary explorations have applied AI-driven pattern recognition across music genres, paving the way for emotion-aware systems, such as wearable devices that trigger mood-matching playlists through AI–big data hybrid frameworks [38].

3. METHODS

OMR algorithms, serving as the cornerstone of intelligent music recognition systems, enable the digitization of musical scores. This technology not only provides an essential prerequisite for in-depth musicological analysis, creative inspiration stimulation, and precise performance realization but also acts as a critical catalyst for advancing the digital transformation of the music industry. The continuous refinement of OMR algorithms not only delivers high-quality data support for intelligent music recognition but also drives sustained innovation in music AI technologies, fostering breakthroughs in accuracy and adaptability.

State-of-the-art OMR methods demonstrate high accuracy in tasks involving single melodies or monophonic structures. However, their performance remains suboptimal when processing complex musical scores featuring har-

monic intervals, block chords, or polyphonic textures. To overcome these limitations, we propose the CNTF model, an autoregressive image-to-sequence neural architecture. Leveraging the hierarchical encoding and decoding capabilities of the transformer, coupled with a balanced sample learning strategy to mitigate data imbalance, the CNTF model significantly enhances the recognition accuracy of structurally intricate musical scores.

3.1. Content Framework

In this section, we delineate the architecture of the CNTF, an autoregressive image-to-sequence neural model proposed herein. The structural components of the model are systematically described through a hierarchical approach. The organizational structure of this section is illustrated in Fig. 1.

3.2. E2E Polyphonic OMR

In the context of OMR, a symbol sequence is an ordered sequence of symbolic tokens representing the musical notation contained in a staff region (e.g., `**ekern.1.0`, `*clefF4`, and `*k[b-e-]`). Each token corresponds to a musical symbol, and the sequence as a whole encodes the notational content to be recognized by the OMR system. State-of-the-art E2E OMR systems frame music transcription as a sequence recognition task. Formally, for a given staff region image x , the objective is to derive the optimal symbol sequence $\hat{s}$. Here, s denotes a candidate symbol sequence and $\hat{s}$ represents the optimal sequence inferred from x :

$$\hat{s} = \underset{s \in \Sigma_a^*}{\operatorname{argmax}} P(s | x), \quad (1)$$

where Σ_a denotes the musical notation vocabulary, i.e., the finite set of atomic notation symbols (such as notes, rests, clefs, and accidentals), and Σ_a^* represents the set of all finite-length sequences composed of these symbols. This

formulation ensures that x is encoded into a syntactically valid sequence of musical symbols.

In practice, the neural network is trained using a connectionist temporal classification (CTC) loss function [39], which models the conditional probability $P(s | x)$ over symbol sequences. The network outputs a probability distribution tensor, where each temporal frame encodes likelihoods for all tokens in the vocabulary Σ_a , including a dedicated blank token. To account for alignment ambiguities (e.g., silent or transitional frames), CTC introduces a blank token ϵ , extending the output vocabulary to $\Sigma'_a = \Sigma_a \cup \{\epsilon\}$. This allows the model to suppress spurious predictions at nonsalient time steps. During inference, the OMR system employs greedy decoding to extract the most probable symbol sequence $\hat{s}$ from the probability tensor. To enforce left-to-right temporal ordering, the model incorporates a vertical compression operation on the feature map. This operation collapses spatial height dimensions into a 1D sequence, ensuring that sequential symbols (e.g., notes and rests) align with their positional encoding in the score. Critically, the vertical compression assumes a left-to-right reading order, consistent with the Western musical notation. For instance, the fourth temporal frame in the compressed feature map consistently encodes features corresponding to the fourth symbol in the sequence, preserving positional fidelity.

The CTC algorithm comprises two phases: training and decoding. During training, CTC computes the marginal probability over all valid alignments between input and output sequences. This requires summing over exponentially many paths, rendering the computation intractable without optimization. To address this, the forward-backward algorithm dynamically computes the likelihood of partial paths. Let $\alpha_{q,t}$ denote the forward variable, representing the probability of being in state q at time t , and $\beta_{q,t}$ the backward variable, which are defined recursively by Eqs. (2) and (3), respectively, enabling the efficient computation of alignment probabilities.

$$\alpha(q, t) = (\alpha(q, t-1) + \alpha(q-1, t-1)) \cdot y_t(l'_q) \quad (2)$$

$$\beta(q, t) = (\beta(q, t+1) + \beta(q+1, t+1)) \cdot y_{t+1}(l'_{q+1}) \quad (3)$$

Here, l denotes the target label sequence and $l' = (l'_1, \dots, l'_Q)$ is the extended sequence with blank symbols inserted between labels and at the ends, where $Q = |l'|$. T denotes the length of the input sequence $X = (x_1, \dots, x_T)$, where each x_t represents the feature vector extracted from the input image (musical score) at time step t . $y_t(k) = P(\text{symbol} = k | X, t)$ is the network output probability of symbol k at time step t . The state index q corresponds to the position in l' , whereas the time index t corresponds to the input frame.

For initialization, we set $\alpha(1, 1) = y_1(l'_1)$, $\alpha(2, 1) = y_1(l'_2)$, and $\alpha(q, 1) = 0$ for $q > 2$.

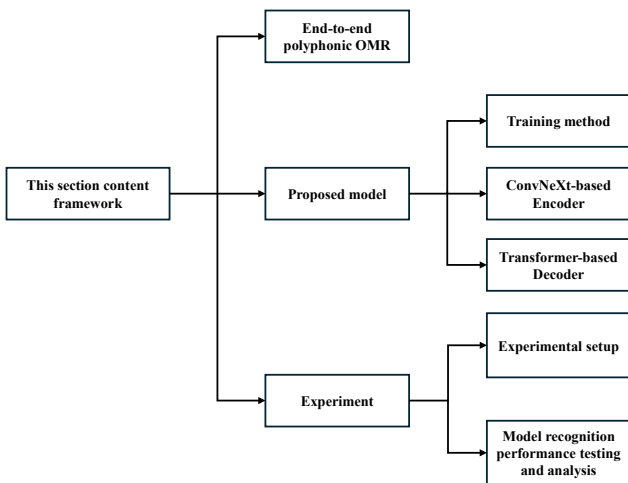


Fig. 1 Content framework of this section.

For termination, the sequence probability is obtained as $P(1 | X) = \alpha(Q, T) + \alpha(Q - 1, T)$.

For backward recursion, we set $\beta(Q, T) = 1$, $\beta(Q - 1, T) = 1$, and $\beta(q, T) = 0$ for $q < Q - 1$.

The CTC loss function, given in Eq. (4), is derived as the negative log-likelihood of the ground-truth sequence, calculated using the forward and backward variables.

$$\mathcal{L}_{CTC}(x, l) = -\log p(l|x) = \sum_{q=1}^Q \frac{\alpha(q, t)\beta(q, t)}{y_t(l'_q)} \quad (4)$$

During decoding, the goal is to find the most probable output sequence by collapsing repeated tokens and removing blank symbols ϵ . For instance, the raw network output “-ab-aa-bb” (where “-” denotes ϵ) is compressed to “abab.” This postprocessing step ensures that the final sequence adheres to the syntactic rules of musical notation, discarding alignment artifacts introduced during training. CTC operates in a fully E2E paradigm, optimizing directly for sequence-level accuracy rather than intermediate alignment steps. This eliminates the need for frame-level annotations, as the loss function evaluates only the final predicted sequence against the ground truth. Given the structural parallels between OMR and speech recognition—both involving sequential data with variable input-output alignment—CTC is well-suited for translating musical score images into symbol sequences, bypassing explicit segmentation of individual symbols.

3.3. Image-to-sequence CNTF Algorithm Modeling via Autoregressive E2E Neural Networks

In this paper, we propose an image-to-sequence framework for transcribing polyphonic musical scores of higher complexity than monophonic compositions. The method leverages an autoregressive E2E neural architecture to generate symbolic transcriptions directly from input score images. The model comprises two core components: (1) an encoder that extracts and encodes features (F_x) from the input score image x , transforming a musical notation into a hierarchical visual-semantic representation; (2) a decoder that autoregressively predicts the probability distribution P_t over symbols at each time step (t) conditioned on F_x and previously generated tokens ($y < t$), following a language modeling framework. The final transcription sequence is derived by iteratively selecting the highest-probability symbol at each time step, thereby achieving robust image-to-symbolic-sequence conversion. The mathematical formulation is as follows:

$$P_t = P(y_t | F_x, (y_0, y_1, y_2, \dots, y_{t-1})), \quad (5)$$

where F_x represents the encoded input feature map derived from the score image, and t indexes the current decoding time step.

$$y_t = \underset{s \in \Sigma}{\operatorname{argmax}} P_t(s), \quad (6)$$

where Σ denotes the symbol vocabulary encoding musical semantics, y_t denotes the predicted symbol at the t -th time step and P_t represents the corresponding probability distribution over the symbol vocabulary Σ . The architectural framework of the CNTF model is illustrated in Fig. 2.

3.4. ConvNeXt-based Encoder

The Swin Transformer (Swin-T) has demonstrated state-of-the-art performance in computer vision tasks, catalyzing extensive research into architectural adaptations. Liu *et al.* [40] systematically modernized conventional CNN backbones (e.g., ResNet50) through architectural refinements such as inverted bottlenecks and large kernel depthwise convolutions, enabling them to emulate the Swin Transformer’s hierarchical modeling capabilities. Our implementation adopts this paradigm: input images are first embedded via ConvNeXt’s stem layer, followed by sequential processing through three ConvNeXt blocks. The resultant features undergo LayerNorm regularization to yield the encoded feature map F_x , which preserves both local texture details and global structural semantics critical for polyphonic score recognition.

Let $x \in \mathbb{R}^{c \times h \times w}$ represent the input image tensor, where h and w denote the spatial dimensions (height and width in pixels respectively), and c corresponds to the number of channels (e.g., $c = 3$ for RGB inputs). The encoder processes x to produce the hierarchical feature map F_x , formally defined as

$$F_x = \operatorname{Enc}(x), \quad (7)$$

where $\operatorname{Enc}(\cdot)$ denotes the ConvNeXt-based encoder.

3.5. Transformer-based Decoder

In this work, we adopt the Transformer decoder [41], which has become the state-of-the-art architecture for variable-length conditional sequence generation tasks. Its effectiveness stems from the multihead attention mechanism’s capability to model long-range temporal dependencies and contextual relationships. The decoder operates as a language model that, at each time step t , generates a probability distribution over the predefined symbol vocabulary Σ , conditioned on the encoder-derived feature map F_x and previously predicted tokens ($\hat{y}_0, \hat{y}_1, \hat{y}_2, \dots, \hat{y}_{t-1}$). The predicted symbol $\hat{y}_t$ is selected to maximize this probability, formalized as

$$\hat{y}_t = \underset{s \in \Sigma}{\operatorname{argmax}} P(\hat{y}_t | F_x, \hat{y} < t). \quad (8)$$

3.6. Class-imbalance Optimization via Focal Loss

Focal Loss [42] offers distinct advantages in addressing class imbalance. Its balancing factor (α) adjusts weights



Figure (a) shows a musical score input and its corresponding ground truth labels. The ground truth labels are as follows:

GROUND TRUTH	
**ekern_1.0	**ekern_1.0
*cleff4	*cleffG2
*k[b-e-]	*k[b-e-]
*M6/8	*M6/8
=	=
4·r	4·F 4·A·- 4·d·- 4·f
8·D·-·L	4·r
8·DD·-·J	
2·r	2·r
=	=
2·r	1·ff 1·aa·- 1·ddd·-
2·DD·-·	
=	=
*_	*_

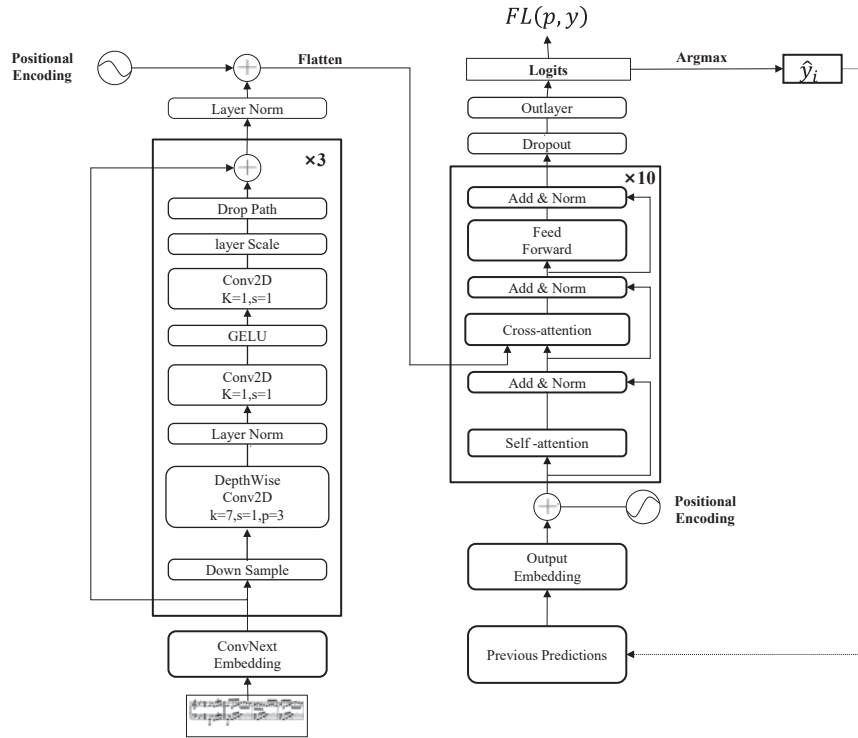
(a) Input test image and ground truth.



Figure (b) shows the model's predicted outputs and their corresponding visual representations. The predicted labels are as follows:

PREDICTION	
**ekern_1.0	**ekern_1.0
*cleff4	*cleffG2
*k[b-e-]	*k[b-e-]
*M6/8	*M6/8
=	=
4·r	4·F 4·A·- 4·d·- 4·f
8·D·-·L	4·r
8·DD·-·J	
2·r	2·r
=	=
2·r	1·ff 1·aa·- 1·ddd·-
2·DD·-·	
=	=
*_	*_

(b) Output sequence and sequence visualization.



(c) Model architecture diagram.

Fig. 2 Figure (a) depicts the input data and their associated ground-truth labels. Figure (b) illustrates the model's predicted outputs and their corresponding visual representations. Figure (c) presents the architectural framework of the CNFT model.

between positive and negative samples, mitigating bias toward majority classes while emphasizing critical minority samples. The focusing parameter (γ) reduces contribu-

tions from easy-to-classify samples and prioritizes hard examples (e.g., small or occluded targets), compelling the model to refine features for challenging cases. Compared

with traditional loss functions, Focal Loss improves training efficiency and accuracy on imbalanced datasets, enhancing detection capability for difficult targets and optimizing model performance.

Focal Loss mitigates performance degradation caused by class imbalance in datasets, addressing both overfitting to majority classes and the underfitting of minority samples. Building upon cross-entropy with a focus on hard examples, Focal Loss is defined as

$$FL(p, y) = - \sum_{c=1}^C \alpha_c (1 - p_c)^\gamma y_c \log(p_c), \quad (9)$$

where C denotes the number of classes, p_c represents the model's predicted probability for class c , y_c is the ground-truth label, α_c is the class-specific balancing factor for class c (adjusting the loss ratio between positive and negative samples), and γ is the focusing parameter that modulates the weighting of easy-to-classify samples versus hard-to-classify samples.

During the initial training phase, we employ the cross-entropy loss function to guide model learning. By training the model on extensive sample data through iterative epochs, it progressively develops foundational classification capabilities, achieving generalized discriminative performance across distinct sample categories.

After undergoing iterative cross-entropy training, the model establishes a foundational classification architecture with baseline accuracy. To further enhance its discriminative performance in challenging polyphonic contexts, we implement Focal Loss for fine-grained optimization.

3.7. Harmonization of Output Formats

The harmonization of output encoding constitutes one of the pivotal components in our system architecture. Among the diverse encodings supported by OMR systems, we prioritized formats with established adoption in digital musicology, such as music encoding initiative (MEI) [43] and MusicXML [44]. Both encodings employ XML-based markup languages to precisely represent musical notation elements and metadata. However, despite this functional sophistication, they exhibit a critical limitation: their XML-based syntax is inherently verbose. This verbosity introduces substantial challenges for OMR systems, as transformations into formats compatible with deep learning architectures frequently incur partial information loss, thereby compromising analytical accuracy and data integrity.

In this study, we adopt the text-based `**kern` encoding format from the Humdrum toolkit [45], hereafter abbreviated as “kern” for conciseness. Among diverse symbolic representations in computational music analysis, the “kern” encoding format stands as one of the most widely adopted standards for encoding musical scores, owing to its

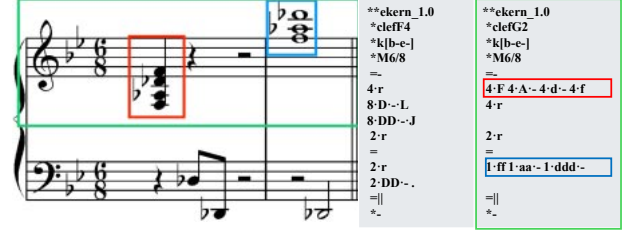


Fig. 3 Example of an excerpt from Humdrum `**kern`-encoded pianoform music.

hierarchical structure and interoperability with analytical workflows. A simple example of an excerpt from Humdrum `**kern`-encoded score6 is shown in Fig. 3.

This approach exhibits notable strengths: first, its vocabulary is characterized by simplicity and intuitiveness, thereby facilitating effortless comprehension and processing; second, the file structure maintains exceptional clarity and ease of parsing. These features render it highly compatible with E2E optical music recognition systems, offering a robust framework to underpin recognition tasks while significantly improving both operational efficiency and accuracy.

Furthermore, “kern” files demonstrate exceptional compatibility, enabling seamless integration with professional music software (e.g., [46,47]) and facilitating efficient data exchange and collaborative workflows. Additionally, “kern” files can be swiftly and automatically transformed into other widely adopted music encoding formats—including the aforementioned MEI and MusicXML—with a single streamlined operation.

3.8. Training Strategy

For the training of both datasets (GrandStaff and Camera GrandStaff), we implement an innovative two-stage strategy. During the first stage, the model is trained using a cross-entropy loss function to establish foundational classification capabilities, ensuring the acquisition of essential discriminative features.

In the second stage, Focal Loss is integrated alongside a sample-balanced learning strategy to enhance the model's adaptability to imbalanced sample distributions, thereby improving its robustness.

Building on the foundational classification capabilities developed in the first stage, the model achieves rapid convergence during the second training phase. As illustrated in Figs. 4 and 5, the loss values for the GrandStaff (110 epochs) and Camera GrandStaff (55 epochs) datasets exhibit a steady decline, confirming efficient convergence.

By applying the sample-balanced learning strategy after establishing the basic feature space, we validated the effectiveness of the phased optimization process. Furthermore, it extends the methodology's applicability to large-

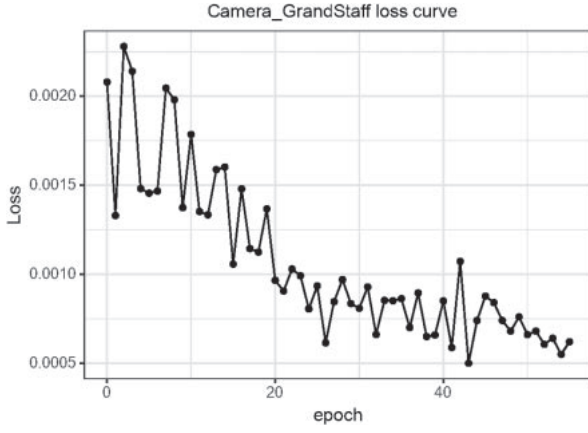


Fig. 4 Camera GrandStaff Stage 2 loss curve.

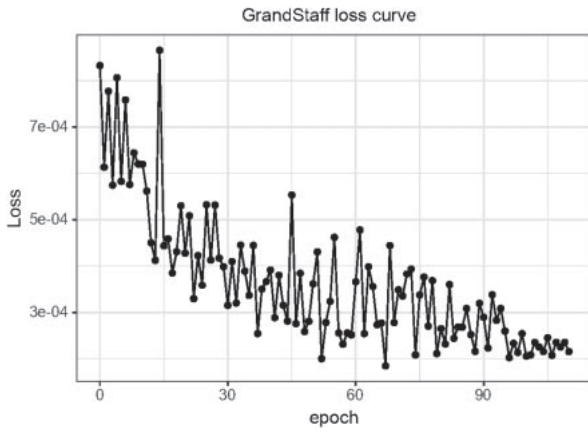


Fig. 5 GrandStaff Stage 2 loss curve.

model fine-tuning scenarios, offering a universal framework for addressing complex data distribution challenges.

4. EXPERIMENTAL SETUP

In this section, we detail the experimental setup, dataset specifications, evaluation metrics, and a comparative analysis of our proposed approach against the method outlined in [29] and other state-of-the-art benchmarks. The hardware and software configurations are summarized in Table 1.

4.1. Dataset Descriptions and Characteristics

To evaluate the efficacy of our model, we performed

Table 1 Environmental setup of the experiment.

OS	ubuntu22.04
CPU model	16 vCPU Intel(R) Xeon(R) Platinum 8474C
System memory	80 GB
GPU	NVIDIA® GeForce RTX™ 4090 D
GPU memory	24 GB

experiments using two publicly available datasets: **GrandStaff** [48] and **Camera GrandStaff**.

GrandStaff [48] is a publicly accessible corpus comprising 53,882 printed piano score images, each representing a single-line musical notation (termed one-line scores) paired with corresponding kern-encoded digital scores.

Structured into two components, the dataset integrates (1) original compositions from six authors sourced from the Humdrum database and (2) synthetic expansions of these compositions. This synthetic augmentation enhances the diversity of musical sequences and structural patterns within the dataset.

The dataset is officially partitioned into a test set (7,000 original scores) and a training set (46,882 synthesized samples). Figure 6 illustrates representative examples of the data.

Camera GrandStaff augments the original corpus by applying synthetic image distortions to simulate low-fidelity captures (e.g., blur, noise, or perspective warping), mimicking real-world photorealistic degradation. Representative samples are shown in Fig. 7.

4.2. Evaluation Metrics

To rigorously evaluate the model's learning capability, we propose three evaluation metrics—informed by prior research methodologies—to quantify its performance:

Character Error Rate (CER) [48]: It quantifies the discrepancy in the smallest semantic units of output encoding by measuring the number of editing operations (e.g., insertions, deletions, and substitutions) required to transform the recognized character sequence into the ground-truth sequence, normalized by the reference sequence's total character length. *CER* is calculated as

$$CER = \frac{Edit_{char}}{N_{ref}}, \quad (10)$$



Fig. 6 Example diagram of GrandStaff's score.



Fig. 7 Illustration of an example score generated by Camera GrandStaff, representing a distorted variant of the notation framework depicted in Fig. 6.

where $Edit_{char}$ denotes the number of edit operations required to align the recognized character sequence with the ground-truth sequence and N_{ref} corresponds to the length of the ground-truth character sequence.

Symbol Error Rate (SER) [48]: It is a widely adopted metric that evaluates errors at the symbol level. SER indicates the number of edit operations (e.g., insertions, deletions, and substitutions) required to align the recognized symbol sequence with the ground-truth sequence, normalized by the length of the reference symbol sequence N_{sym} . Its formula is

$$SER = \frac{Edit_{sym}}{N_{sym}}, \quad (11)$$

where $Edit_{sym}$ denotes the number of edit operations (e.g., insertions, deletions, and substitutions) required to align the recognized symbol sequence with the ground-truth sequence and N_{sym} corresponds to the length of the reference (ground-truth) symbol sequence.

Line Error Rate (LER) [48]: It is a sequence-level metric quantifying the number of edit operations (e.g., insertions, deletions, and substitutions) required to align the recognized line content with the ground-truth line, normalized by either the reference line length N_{line} or the total element count. Its formula is

$$LER = \frac{Edit_{line}}{N_{line}}, \quad (12)$$

where $Edit_{line}$ denotes the number of edit operations (e.g., insertions, deletions, and substitutions) required to align the recognized line with the ground-truth line and N_{line} represents the total number of elements in the reference line.

4.3. Analysis of Class Balance in the Dataset

To analyze the class imbalance in our dataset, we present the detailed class distributions of the GrandStaff dataset. Note that the Camera GrandStaff dataset is a distortion-introduced version of GrandStaff, and therefore, its class distribution is identical; for brevity, only the GrandStaff statistics are shown. As summarized in Tables 2 and 3, the top 10 classes account for approximately 40% of all samples, whereas the remaining classes (9,434 types in the test set and 18,078 types in the training set) constitute roughly 60% of the dataset. Notably, the most frequent class (“.”) alone comprises about 19–20% of the samples, whereas the 10th-ranked class (“*v”) only accounts for around 1%.

These statistics clearly demonstrate a pronounced class imbalance, with a small number of classes dominating the dataset, whereas the majority of classes have relatively few samples. Such imbalance poses a challenge for standard loss functions, which tend to bias models toward the

Table 2 Class distribution in the GrandStaff test dataset.

Class	Count	Proportion (%)
.(1st)	185,032	19.79
=(2nd)	78,234	8.36
*v(10th)	10,668	1.14
Top 10 total	376,836	40.31
Other classes (9,434 types)	557,949	59.69

Table 3 Class distribution in the GrandStaff training dataset.

Class	Count	Proportion (%)
.(1st)	985,486	19.70
=(2nd)	418,167	8.36
*v(10th)	56,916	1.13
Top 10 total	2,013,607	40.26
Other classes (18,078 types)	2,987,687	59.74

majority classes. To address this, we employ the Focal Loss function, which down-weights the loss contribution of well-classified examples and focuses more on hard-to-classify and minority-class samples, thereby improving model performance on underrepresented classes.

5. RESULTS AND ANALYSIS

All baseline models in our experiments are implemented following the E2E OMR framework, which is, to the best of our knowledge, the only published E2E OMR framework extending beyond monophonic transcription with ad-hoc adaptations. The publicly available dataset used in this study is also sourced from Ríos-Vila *et al.* [48]. We adopt the three models proposed by Ríos-Vila *et al.* [48] for benchmarking: (1) CNNT, which replaces the recurrent layers in the CRNN framework with a Transformer-based encoder–decoder module to enhance global context modeling; (2) FCN, a fully convolutional encoder-only network performing pixel-wise feature extraction and symbol classification; and (3) the standard CRNN, which combines convolutional feature extraction with a recurrent decoder for sequential modeling. For all baseline models, we follow the dataset partition, training settings, and optimization strategies described by Ríos-Vila *et al.* [48], including task-specific loss functions, optimizer configurations, learning rates, and early stopping criteria. These measures ensure that the baseline results are reliable, reproducible, and directly comparable to our proposed method.

In addition to the baseline model, our benchmark comparisons include three advanced architectures from Ríos-Vila *et al.* [29]: (i) SMT_{CNN} , (ii) SMT_{SWIN} , and (iii)

Table 4 Performance comparison of models on the GrandStaff and Camera GrandStaff datasets for CER, SER, and LER (%).

Model	GrandStaff			Camera GrandStaff		
	CER	SER	LER	CER	SER	LER
Baseline						
FCN	14.6	23.9	67.9	20.6	30.2	69.0
CRNN	5.0	7.3	23.2	7.2	9.9	29.5
CNNT	7.9	11.1	32.4	9.4	12.3	33.3
SMT_{CNN}	5.7	7.6	19.5	6.9	8.5	20.2
SMT_{SWIN}	53.2	70.1	98.2	60.3	82.3	100.0
SMT_{NextT}	3.9	5.1	13.1	5.3	6.2	13.5
CNFT	2.3	3.2	9.6	3.4	4.4	11.7

SMT_{NextT} . These models have demonstrated superior performance in prior studies.

We present the quantitative evaluation results of our proposed model in Table 4, including performance metrics on both the Camera-GrandStaff (augmented subset) and standard GrandStaff benchmark datasets. For a fair comparison, all experiments in our study strictly followed the same dataset partition and experimental settings by Ríos-Vila *et al.* [29], where the training set consists of 46,882 synthesized scores and the test set contains 7,000 original scores. The SMT results shown in Table 4 were reproduced using the official checkpoints provided by Ríos-Vila *et al.* [29] under this configuration, ensuring consistency and comparability across all reported results.

Our proposed model demonstrates state-of-the-art performance across all evaluation metrics on both the Camera-GrandStaff and GrandStaff datasets. Compared with the top-performing existing model, SMT_{NextT} , our approach achieves a consistent performance gain of approximately 2 percentage points in every metric category.

6. DISCUSSION

This study introduces CNTF, an E2E Transformer-based framework designed to overcome the limited accuracy of conventional OMR systems in processing complex musical scores, including polyphonic and multivoiced compositions. The model integrates a ConvNeXt encoder with a Transformer decoder and employs Focal Loss optimization, achieving superior performance in transcribing structurally intricate scores. This architecture combines ConvNeXt's spatial feature extraction with Transformer's sequential modeling capabilities, enhanced by Focal Loss training, thereby achieving robust accuracy in complex music transcription tasks.

6.1. Model Innovation and Performance Advantages

The core innovation of CNTF lies in its E2E image-to-

sequence architecture, which eliminates the limitations of traditional multistage OMR methods (e.g., isolated image preprocessing and symbol recognition steps). The regression mechanism effectively handles long-range dependencies in polyphonic sequences and demonstrates enhanced modeling capabilities for temporally complex elements such as harmonic intervals and chords. Experimental results indicate that CNTF outperforms baseline models (CRNN and SMT series) with 2–3 percentage point improvements in CER, SER, and LER. Additionally, the model exhibits superior robustness to low-quality musical images with distortion noise in the Camera GrandStaff dataset, confirming its adaptability to real-world scenarios.

6.2. Research Value and Practical Significance

CNTF offers an efficient framework for the automated transcription of complex scores while advancing OMR from monophonic melody analysis to polyphonic and multivoiced music recognition. Its applications span music education (e.g., AI-driven score proofreading), cultural heritage preservation (e.g., digitizing medieval neume manuscripts), and music production (e.g., AI-assisted orchestration). The optimized E2E architecture further validates Transformer's versatility in 2D image-to-sequence tasks, establishing a paradigm for cross-modal music processes, such as audio-score synchronization and symbolic score generation. Future advancements in model compression and format interoperability could position OMR as a critical nexus between physical scores and digital music ecosystems, facilitating the intelligent preservation and innovation of musical artistry.

7. CONCLUSION

Conventional OMR systems exhibit limited accuracy when processing structurally complex musical notations, particularly polyphonic scores. To address this issue, we propose a Transformer-based optimization framework that integrates a ConvNeXt encoder for hierarchical feature

extraction and a Transformer decoder for sequential dependency modeling. This architecture provides a more effective representation of intricate musical patterns. Additionally, to mitigate class imbalance and enhance recognition robustness for under-represented notation classes, we employ Focal Loss as the training objective.

The effectiveness of this model is demonstrated through experiments on two publicly available polyphonic piano score datasets, GrandStaff and Camera GrandStaff. Moreover, a comparative analysis with existing state-of-the-art models shows that our method consistently outperforms prior approaches in sheet music recognition performance, with notable improvements in key metrics such as note recognition accuracy and complex typesetting adaptability.

While our model achieves strong performance, several limitations warrant further investigation. First, its reliance on single-format output limits compatibility with diverse real-world application scenarios. Second, the high computational complexity and parameter count hinder deployment in resource-constrained edge devices. Future work will prioritize (1) architectural optimizations to reduce computational and memory overhead through lightweight network design, and (2) enhancing model generalizability to achieve compatibility with diverse musical formats (e.g., lead sheets and tablature) and stylistic variations. These advancements are critical for broadening the practical adoption of OMR systems across music education, digitization, and archival workflows.

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